

DETERMINISTIC CONVERGENCE LIFECYCLE

Ex-Mortem Blame to Convergence Intelligence.

Used-loop 5-stage workflow that deterministically proves software before, during, and after promotion.

02

UNDERSTAND

Isolate unvalidated version combinations

03

REHEARSE

Simulate promotions counterfactually

04

PROVE

Execute isolated compatibility probes

05

REMEDiate

Draft backwards-compatible shims

CONSTRUCT — Ground Truth

NON-INVASIVE READ-ONLY

git commit tags or release manifests, Meridian deploys non-mutating Kubernetes clusters, OpenShift pods, and container registries to discover actively serving traffic.

Digests,

DETERMINISTIC VERDICT

SAFETY GUARANTEE

Verified Environment Reality Matrix

Zero cluster mutation, strictly read-only

ARCHITECTURE • DETERMINISTIC CORE + IBM BOB 2.0 AGENT SWARM

Dual-Engine Architecture: Mathematical Rigor + Agentic Collaboration.

Meridian pairs a deterministic Python reconciliation engine with IBM Bob 2.0's autonomous specialist agents via 17 custom Model Context Protocol (MCP) tools.

FOUNDATION LAYER

Deterministic Python Engine

Release validation decisions must be mathematically verifiable. The core reconciliation engine runs zero LLM calls in its verdict path—preventing hallucinations and ensuring 100% reproducible results for auditors and CABs.

Release Investigator Agent

Parses git commits, PR diffs, and CAB tickets (CR-4471) to detect scope.

Environment Investigator Agent

Queries live pods across DEV, TEST, STAGE, PROD to uncover actual running versions.